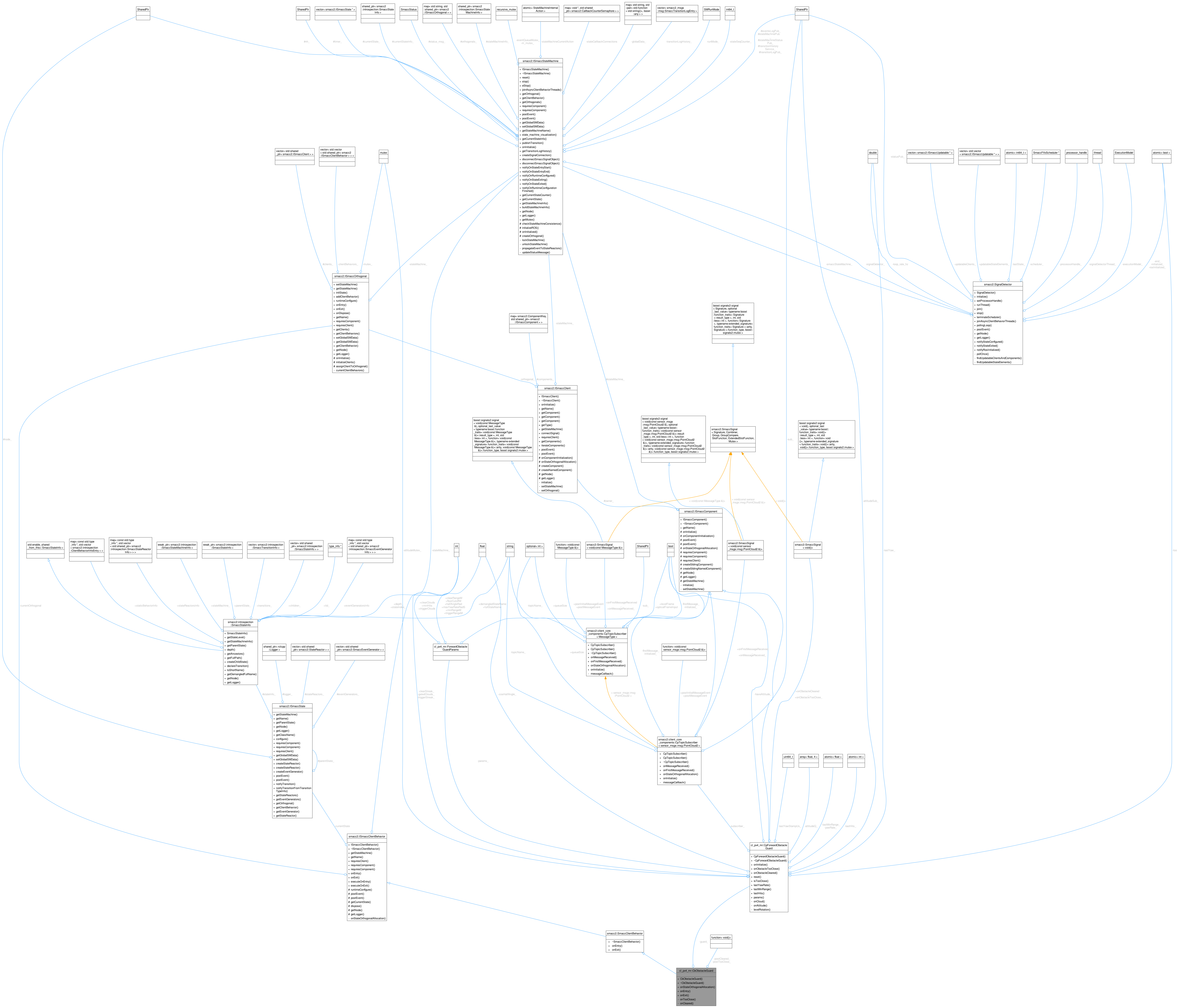
SharedPtr
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2 ::introspection::SmaccState Info >
SmaccStatus
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
recursive_mutex
atomic< StateMachineInternal Action >
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
#nh_
#timer_
#currentState_
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
-eventQueueMutex_ -m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#eventsLogPub_ #stateMachinePub_ #stateMachineStatus Pub_ #transitionHistory Service_ #transitionLogPub_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
double
-statusPub_
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
atomic< int64_t >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
atomic< bool >
-smaccStateMachine_
-signalDetector_
-loop_rate_hz
-updatableClients_
-updatableStateElements_
-lastState_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-end_ -initialized_ -rosInitialized_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
vector< std::shared _ptr< smacc2::ISmaccClient > >
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
mutex
#clients_
-clientBehaviors_
-mutex_
-stateMachine_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
-stateMachine_
-orthogonal_
#components_
#stateMachine_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
boost::signals2::signal < void(const MessageType &), optional_last_value < typename boost::function _traits< void(const MessageType &)>::result_type >, int, std ::less< int >, function< void(const MessageType &)>, typename extended _signature< function_traits< void(const MessageType &)>::arity, void(const MessageType &)>::function_type, boost::signals2::mutex >
boost::signals2::signal < void(const sensor_msgs ::msg::PointCloud2 &), optional _last_value< typename boost:: function_traits< void(const sensor _msgs::msg::PointCloud2 &)>::result _type >, int, std::less< int >, function < void(const sensor_msgs::msg::PointCloud2 &)>, typename extended_signature< function _traits< void(const sensor_msgs::msg::PointCloud2 &)>::arity, void(const sensor_msgs::msg::PointCloud2 &)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
-sub_
-attitudeSub_
-lastYaw_
-tooClose_
#owner_
< void(const MessageType &)>
< void(const sensor _msgs::msg::PointCloud2 &)>
< void()>
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
#node_
-currentOrthogonal
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
weak_ptr< smacc2::introspection ::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection ::SmaccStateInfo >
vector< smacc2::introspection ::SmaccTransitionInfo >
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
type_info *
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
int
float
string
optional< int >
function< void(const MessageType &)>
smacc2::SmaccSignal < void(const MessageType &)>
SharedPtr
bool
smacc2::SmaccSignal < void(const sensor _msgs::msg::PointCloud2 &)>
smacc2::SmaccSignal < void()>
+staticBehaviorInfo
+stateReactorsInfo
+stateMachine_
+parentState_
+transitions_
+children_
+tid_
+eventGeneratorsInfo
+depth_ +stateIndex_
-attitudeMutex_
-stateMachine_
+clearClouds +minHits +triggerClouds
+clearRangeM +floorCutoffM +halfAngleRad +maxYawRateRadS +minRangeM +triggerRangeM
+demangledStateName +fullStateName
-topicName_
+queueSize
+postInitialMessageEvent +postMessageEvent
+onFirstMessageReceived_ +onMessageReceived_
-sub_
+levelFrame +opticalFrameInput
-firstMessage_ -initialized_
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::StateReactor > >
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
cl_px4_mr::ForwardObstacle GuardParams
smacc2::client_core _components::CpTopicSubscriber < MessageType >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
function< void(const sensor_msgs::msg::PointCloud2 &)>
#stateInfo_
#logger_
#stateReactors_
#eventGenerators_
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
-clearStreak_ -gatedClouds_ -triggerStreak_
-cosHalfAngle_
-topicName_
+queueSize
-firstMessage_ -initialized_
< sensor_msgs::msg ::PointCloud2 >
+postInitialMessageEvent +postMessageEvent
+onFirstMessageReceived +onMessageReceived_
-haveAttitude_
+onObstacleCleared +onObstacleTooClose_
smacc2::client_core _components::CpTopicSubscriber < sensor_msgs::msg::PointCloud2 >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
uint64_t
array< float, 4 >
atomic< float >
atomic< int >
-params_
-subscriber_
-lastYawStampUs_
-attitudeQ_
-lastMinRange_ -yawRate_
-lastHits_
-currentState
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
cl_px4_mr::CpForwardObstacle Guard
+ CpForwardObstacleGuard()
+ ~CpForwardObstacleGuard()
+ onInitialize()
+ onObstacleTooClose()
+ onObstacleCleared()
+ reset()
+ isTooClose()
+ lastYawRate()
+ lastMinRange()
+ lastHits()
+ params()
- onCloud()
- onAttitude()
- levelRotation()
smacc2::SmaccClientBehavior
+ ~SmaccClientBehavior()
+ onEntry()
+ onExit()
-guard_
function< void()>
-postCleared_ -postTooClose_
cl_px4_mr::CbObstacleGuard
+ CbObstacleGuard()
+ ~CbObstacleGuard()
+ onStateOrthogonalAllocation()
+ onEntry()
+ onExit()
- onTooClose()
- onCleared()